

RESEARCH INTEREST

Spatial Intelligence, Neural Representation, Multi-Agent Systems, Field Robotics

EDUCATION

University of Michigan

Ph.D. Candidate in Robotics

Aug. 2025 –

- Advisor: Prof. Yulun Tian

University of Tokyo

Apr. 2023 – Mar. 2025

M.E.S. in Human & Engineered Environmental Studies

- Advisor: Prof. Atsushi Yamashita
- Thesis: Switching-based Multi-modal SLAM for Extreme and Degraded Environments

University of Osaka

Apr. 2017 – Mar. 2023

B.E. in Mechanical Engineering

(Military Service included)

- Advisor: Prof. Masamitsu Kurisu
- Thesis: LiDAR-visual SLAM for Online Mapping of Unpaved Road Surface

PREPRINT

- [1] **LatentAM: Real-Time, Large-Scale Latent Gaussian Attention Mapping via Online Dictionary Learning**

[Junwoon Lee](#), Yulun Tian

Available on arXiv, [arXiv:2602.12314](#) [Link]

PUBLICATIONS

- [1] **CubeDVO: Cubemap-Spherical Deep Visual Odometry for a Monocular 360-Degree Camera**

Taisei Ando, [Junwoon Lee](#), Takuya Igaue, Kohtaro Nakamura, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
IEEE Robotics and Automation Letters (RA-L), 2026.

- [2] **Robot Localization by Data Integration of Multiple Thermal Cameras in Low-light Environment**

Masaki Chino, [Junwoon Lee](#), Qi An, Atsushi Yamashita

International Journal of Automation Technology, 2025. [Link]

- [3] **Efficient Distortion Mitigation in Equirectangular Images for Two-View Pose Estimation**

Taisei Ando, [Junwoon Lee](#), Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita

International Journal of Automation Technology, 2025. [Link]

- [4] **Self-TIO: Thermal-Inertial Odometry via Self-supervised 16-bit Feature Extractor and Tracker**

[Junwoon Lee](#), Taisei Ando, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita

IEEE Robotics and Automation Letters (RA-L), 2025. Presented at IROS'25. [Link]

- [5] **TC-LTIO: Tightly-coupled LiDAR Thermal Inertial Odometry for LiDAR and Visual Odometry Degraded Environments**

[Junwoon Lee](#), Taisei Ando, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita

International Conference on Control, Automation and Systems (ICCAS), 2024. [Link] (Best Paper Award)

- [6] **Highly Accurate and Fast Two-view Pose Estimation by Fast Reduction of Spherical Image Distortion Effects**

Taisei Ando, [Junwoon Lee](#), Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita

International Conference on Control, Automation and Systems (ICCAS), 2024. [Link]

- [7] **Switch-SLAM: Switching-Based LiDAR-Inertial-Visual SLAM for Degenerate Environments**
Junwoon Lee, Ren Komatsu, Mitsuru Shinozaki, Toshihiro Kitajima, Hajime Asama, Qi An, Atsushi Yamashita
IEEE Robotics and Automation Letters (RA-L), 2024. Presented at ICRA@40. [\[Link\]](#)
- [8] **Three-dimensionalized Feature-Based LiDAR-visual Odometry for Online Mapping of Unpaved Road Surface**
Junwoon Lee, Masamitsu Kurisu, Kazuya Kuriyama
Journal of Field Robotics, 2024. [\[Link\]](#)

RESEARCH EXPERIENCE

- Research Assistant**, University of Michigan *Aug. 2025 –*
 Scalable Spatial Intelligence Lab.
 - Developing a VLM features-embedded Gaussian Splatting SLAM system.
 - Developing a LiDAR-visual-GNSS-inertial Multi-robot experimental platform.
- Research Assistant**, University of Tokyo *Apr. 2023 – Jul. 2025*
 Real World Robot Informatics Lab.
 - Developed a multi-modal SLAM system for complex scenes.
 - Developed a self-supervised point tracker for thermal inertial odometry.
- Research Assistant**, University of Osaka *Apr. 2022 – Mar. 2023*
 Komatsu MIRAI Construction Equipment Cooperative Research Center
 - Suggested an intensity-weighted point cloud registration.
 - Developed a mapping system for an unpaved road surface.

HONORS AND AWARDS

- Department Fellowship** *Sep. 2025 – Aug. 2026*
 - Fellowship for 1st-year doctorate studies.
- BOOST NAIS AI Fellowship (declined)** *Apr. 2025 – Mar. 2028*
 - Full Fellowship (Ph.D.) at UTokyo, \$75,000+ USD
- Dean's Award** *Mar. 2025*
 - Graduated at the top of the department (1/39)
- Best Paper Award** *Oct. 2024*
 - ICCAS'24 (1/400)
- IEEE RAS Travel Grant** *Sep. 2024*
 - ICRA@40, \$2,000 USD
- Rotary Yoneyama Memorial Foundation Scholarship** *Apr. 2023 – Mar. 2025*
 - Full scholarship (M.S.), \$30,000+ USD
- Korea-Japan Joint Government Scholarship** *Apr. 2017 – Mar. 2023*
 - Full scholarship (B.S.), \$75,000+ USD

TEACHING

- Teaching Assistant**, UTokyo FEN-SC3102S1 Exercises for Mathematics 2C *Apr. 2024 – Jul. 2024*

MENTORING

- Siddhant Bohra (B.S., University of Michigan), Full stack mobile robot-system implementation *May. 2026 –*
 Arnav Shah (M.S., University of Michigan), Foundation model-based loop closure *Feb. 2026 –*
 Yiyu Wang (B.S., University of Michigan), Sensor suite design, SLAM benchmarking *Sep. 2025 –*
 Kohei Yamaguchi (B.S., University of Tokyo), Learning-based point cloud registration *Apr. 2025 –*
 Taisei Ando (Ph.D./M.S./B.S., University of Tokyo), Learning-based omnidirectional SLAM *Apr. 2023 –*

SERVICES

Academic Reviewer

- T-RO (2025), RA-L (2024–2026), ICRA (2025–2026), IROS (2025–2026), T-Mech (2026), T-ASE (2024–2025), Journal of Field Robotics (2025), IEEE Sensors Journal (2024–2025), ICCAS (2024), Scientific Reports (2024)

Sergeant, Republic of Korea Army

Apr. 2020 – Oct. 2021

- Served as a frontline guardian at a coastline observation post in the 23rd Security Brigade

SKILLS

Research Skills

- Program Languages : C/C++, Python
- Professional : ROS1, ROS2, GTSAM, Ceres Solver, OpenCV, PyTorch, TensorRT, Git, 3D CAD

Languages: English (Professional), Japanese (Professional), Korean (Native)

PATENT

1. Kaoru Adachi, Masamitsu Kurisu, Junwoon Lee, “Terrain Detection System and Method,” *Japanese Patent app:2023-105215 / open:2025-005158*, Filed on June 27, 2023.

REFERENCES

References available upon request.